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**BSCS – 3B**

**Performance Task 1: From Data to Prediction – Analyzing Real-World Machine Learning Application**

### **Chosen Machine Learning Application**

**Image Recognition** - A system that uses Machine Learning to recognize what is shown in a picture. It can identify different objects or things in an image by learning from many examples.

### **Problem Description**

Many people and organizations need a faster and more accurate way to identify objects in images. Manually checking and identifying images can take a lot of time, especially when there are many images to process. I wish to train a model with the capability to learn visual patterns from different images so that it can recognize and classify objects automatically.

### **Part A - Identify the ML Problem**

#### **1. What is the application?**

Image Recognition

#### **2. What problem are you trying to solve?**

The problem is how to automatically identify or recognize objects in an image without checking each image manually.

#### **3. What are the inputs?**

The inputs are images or pictures that contains different objects.

#### **4. What is the expected output?**

The expected output is the name or classification of the object found in the image.

#### **5. What patterns or relationships might the ML system learn?**

The system can learn patterns from the shape, color, size, and other features of objects in different images.

#### **6. What data would the system need?**

The system would need many images with labels, such as pictures labeled as cat, dog, car, person, and other objects.

### **Part B – Traditional Programming vs. Machine Learning**

#### **Traditional Programming:**

Input -> Algorithm -> Output

For image recognition the programmer would need to create specific rules to identify objects in an image. For example, the programmer may create rules based on the shape, color, or size of an object. This can be difficult because there are many different appearances of the same object.

### **Machine Learning:**

#### **Input + Output/Data -> Machine Learning Algorithm**

In Machine Learning we give the system many images together with their correct labels. The algorithm will study the images and learn the patterns from them. After that, it can use what it learned to recognize objects in a new image without making every rule manually.

#### **Part C - Create Your Own Mini Dataset**

| <b>Input</b>              | <b>Output</b> |
|---------------------------|---------------|
| Red round fruit           | Fruit         |
| Yellow curved fruit       | Fruit         |
| Yellow diamond shape sign | Traffic Sign  |
| Long ears and small body  | Animal        |
| Orange round fruit        | Fruit         |

#### **1. Identify the pattern**

The system looks at the visual features of the image and identifies what category the object belongs to.

#### **2. Determine the missing output**

For example, if the input is “Orange, round fruit” the output would be Fruit.

#### **3. Write the mathematical Relationship/function**

Visual features of image -> Object category

#### **4. Explain how a machine learning algorithm could learn this relationship from the example.**

The ML algorithm can learn from many images with their correct categories. It studies features like color, shape, size, and appearance, then learns which category matches those features. After learning from enough examples, it can classify a new image into the correct category.

#### **Part D - Explain the Training Process**

A Machine Learning algorithm learns by looking at the inputs and the correct outputs from the data. It studies the features of the images like the color, shape, and appearance to know what object it is. The more examples the system gets, the more it can learn the pattern between the image and its correct output. After training, it can use what it learned to predict the object in a new image.

## **Reflection**

### **Why is learning from data useful instead of manually programming every possible rules?**

Learning from data is useful because the system can learn different patterns without the programmer making a rule for every possible situation. It can also handle many different examples, which is hard to do manually. For image recognition, there are many kinds of images so making rules one by one can take a lot of time. I think learning from data makes the system more flexible and easier to improve when there are new examples.